

$$m_i = \frac{F_i - F_{0,i}}{F_{0,i}} \Rightarrow \frac{F_i}{F_{0,i}} = 1 + m_i \quad \begin{array}{l} \text{primary: } i=1 \\ \text{secondary: } i=2 \end{array}$$

$$t_i = \frac{F_i}{F_{0,i}}$$

$\Rightarrow$ Combining modulation + transit:

$$\frac{F_i}{F_{0,i}} = (1 + m_i) t_i$$

Combining primary & secondary:

$$F = F_1 + F_2 = (1 + m_1) t_1 F_{0,1} + (1 + m_2) t_2 F_{0,2}$$

$$\text{Let } f_2 = \frac{F_{0,2}}{F_{0,1}}$$

$$F_{0,2} = f_2 F_{0,1}$$

$$F_0 = (1 + f_2) F_{0,1}$$

$$F = [(1 + m_1) t_1 + f_2 (1 + m_2) t_2] F_{0,1}$$

$$\Rightarrow \frac{F}{F_0} = \frac{(1 + m_1) t_1 + f_2 (1 + m_2) t_2}{1 + f_2}$$

$$\frac{F}{F_0} = \frac{(1+m_1)t_1 + f_2(1+m_2)t_2}{1+f_2}$$

For fitting eBEER reflection & beaming:

$$m_1 = m_{e,1} + \alpha_{r,1} m_{r,1} + \alpha_{b,1} m_{b,1}$$

$$m_2 = m_{e,2} + \alpha_{r,2} m_{r,2} + \alpha_{b,2} m_{b,2}$$

$$\Rightarrow (1+f_2) \frac{F}{F_0} - (1+m_{e,1})t_1 - f_2(1+m_{e,2})t_2 =$$

$$\alpha_{r,1} m_{r,1} t_1 + \alpha_{b,1} m_{b,1} t_1 + \alpha_{r,2} f_2 t_2 m_{r,2} + \alpha_{b,2} f_2 m_{b,2} t_2$$

$$\text{RHS} = (1+f_2) \frac{F}{F_0} - (1+m_{e,1})t_1 - f_2(1+m_{e,2})t_2$$

$$A = \begin{pmatrix} | & | & | & | \\ m_{r,1} t_1 & f_2 m_{r,2} t_2 & m_{b,1} t_1 & f_2 m_{b,2} t_2 \\ | & | & | & | \end{pmatrix}$$

With

ω : arg of periapsis

ν : true anomaly

Superior conj when $\omega + \nu = 90^\circ$

Procedure for finding initial walker positions

1. Circular v.s. eccentric:

If masked BLS is significant $e > 0$

If not signif and odd odd & even depths ^{same}
consider two cases: $M_1 = M_2, e = 0$ or $e > 0$ single elix

2. Get each combination Π_i of the following

$$\log_{10}(t) \in \{6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10\}$$

$$\left[\frac{F_2}{H}\right] \in \{-1, -0.5, 0, 0.5\}$$

$$\nexists e \approx 0: e \in \{0, 7 \times \mathcal{U}(0, 0.2)\}$$

for each e choose $\omega = \mathcal{U}(0, 360^\circ)$

$$\nexists e > 0: \omega \in \{0, 45^\circ, 90^\circ, \dots, 315^\circ\}$$

For each ω solve for e
to match phase difference b/w
unmasked and masked transit
times.

3. For each Π_i above find M_1, M_2, i as follows:

follows:

- Isolate LC points near deeper eclipse (LC_d)

- Isolate LC points near shallower eclipse (LC_s)

- single eclipse case handled later

- Define function to optimize i given M_1, M_2 :
 $(M_1, M_2) \rightarrow (i, \log \text{ likelihood})$

a) Create LC with $i=0$ to see which eclipse is deeper & set eclipse time accordingly

b) Find i by optimizing log-likelihood of LC_1 . This tunes i to match eclipse depths of deeper eclipses

- Wrap function around the above to tune M_2 holding M_1 fixed to maximize log-likelihood of LC_2 .

- Wrap function around the above to maximize SED log-likelihood by tuning M_1

This should result in 256 combinations of:
 $P_{orb}, \log_{10} t, \left[\frac{F_2}{H}\right], e, \omega, i, M_1, M_2, t_0$
leaving only out-of-eclipse parameters & systematic uncertainties.